Twelve forward predictions: Paper 4 Table S4

ID / tier	Prediction	Target	Evidence against	Horizon / current status
P1 / T3	Nicotinic agonist lowers effective threshold	Threshold shift d ≥ 0.4; P3b onset ~20–30 ms earlier	Equivalence test (TOST) bounds threshold shift within d = ±0.2	Near-term (1–2 yr); Indirect support from nicotinic-attention literature; untested in ignition paradigm
P2 / T3	Near-threshold bistability with critical slowing	Critical-slowing signature; ΔAIC > 10 (bistable vs graded)	Continuous graded relationship	Near-term (1–2 yr); Partial (bistable-perception literature); slowing not tested for ignition
P3 / T3	200–300 ms frontoparietal TMS abolishes report, spares priming	Report-accuracy drop d ≥ 0.5 with priming intact	Both disrupted equally, or access persists	Near-term (2–3 yr); Partial (TMS-masking literature); specific window untested
P4 / T3	Ignition margin $S_t - \theta_eff,_t$ predicts access at $d' \geq 0.5$	$d' \geq 0.5$; AUC ≥ 0.70; partial $R^2 \geq 0.05$ (Π^e, Π^i_eff each)	$d' < 0.5$ under the §7.4 design, or a stimulus-intensity-only model matches within $\Delta d'$ 0.05	Near-term (1–2 yr); Specified; simulation-based parameter recovery only (Pesochin, 2026a); prospective test pending
P5 / T2	Attentional cueing raises stimulus–response MI I(S; R) by ≥ 0.3 bits (S = cue-congruent target identity; R = report; Δ, H(S), baseline I(S; R) and the bias-corrected estimator to be fixed before registration), with Δ < H(S) − baseline I(S; R)	ΔI(S; R) ≥ 0.3 bits, with 0.25 < Δ < H(S) − baseline I(S; R)	Equivalence test bounds ΔMI within ±0.25 bits	Mid (2–3 yr); Untested; consistent with attention-gain literature
P6 / T2	Conscious bandwidth asymptotes within 10–40 bits/s	Asymptote within 10–40 bits/s, modality-invariant	Modality-invariant asymptote outside 10–100 bits/s, or modality variation exceeding the 10–40 bits/s range	Mid (2–4 yr); Partial (classical bandwidth estimates); not tested as ignition asymptote
P7 / T2	θ_t approximates Neyman–Pearson optimum	Deviation ≤ 2 SD from optimum	Deviation > 2 SD	Mid (2–3 yr); Untested
P8 / T2	Interoceptive TE to report ≥ 0.05 bits/trial, exteroceptive TE partialed	TE_intero ≥ 0.05 bits/trial (partial); TE_intero > TE_extero exploratory	90% CI of surrogate-corrected TE_intero wholly below 0.01 bits/trial (equivalence test; §7.4)	Mid (3–4 yr); Untested; motivated by interoception literature
P9 / T1	Ignition trials cost more frontoparietal energy	≥ 5% ^{31}P-MRS metabolic difference; d ≥ 0.4, N ≥ 52 within-subject. BOLD contrast used for region selection only	Equivalence test bounds differential within ±2% after controls	Long (5–10 yr); Unmeasured — primary Tier 1 gap
P10 / T1	Threshold-gated more energy-efficient than continuous	Energy saving exceeding the between-architecture variance of the matched-budget comparison in §7.5.2; no point target is registered	Continuous matches or beats it	Mid–long (3–5 yr); Not modeled; design specified (§7.5.2)
P11 / T1	Metabolic depletion elevates θ_t	Threshold elevation d ≥ 0.4	Equivalence test bounds shift within d = ±0.2, or threshold lowered	Mid (3–5 yr); Untested; indirect support from vigilance/fatigue work
P12 / T1	A matched event-energy/event-information experiment yields an empirical κ estimate under the registered calibration convention	κ estimable with uncertainty; [10, 10^3] consistent, (10^3, 10^4] requires explanation	Replicated calibrated κ estimates in ≥ 2 independent datasets/paradigms have preregistered uncertainty intervals wholly outside [10, 10^4] ATP/bit	Long (5–10 yr); Unmeasured; prerequisite calibration not yet performed

Paper 4 exposition IDs differ from Protocol Pred IDs. Pending registration and calibration are retained.